

Research paper

Neural correlates of early- and late-onset nonsuicidal self-injury: Distinctions in resting-state functional connectivity

Hyeri Moon^{a,1}, Jinhee Kim^{a,1}, Gyumyoung Kim^a, Hackjin Kim^a, Taekwan Kim^b, Ji-Won Hur^{a,*}

^a School of Psychology, Korea University, South Korea

^b Mental Health Neuroscience Department, Division of Psychiatry and Max Planck UCL Centre for Computational Psychiatry and Ageing Research, Queen Square Institute of Neurology, University College London, United Kingdom

ARTICLE INFO

Keywords:

Nonsuicidal self-injury
Emotion regulation
Emotion dysregulation
Age of onset
Neuroimaging
Resting-state functional magnetic resonance imaging

ABSTRACT

Background: Nonsuicidal self-injury (NSSI) is a major global health concern, yet the neurophysiological significance of the age of onset remains poorly understood. This cross-sectional study (2019–2022) examined differences in resting-state functional connectivity (rsFC) among young Korean adults (aged 19–29) with early-onset NSSI, late-onset NSSI, and healthy controls, with a focus on emotion regulation networks.

Methods: Following screening of 215 individuals, 91 unmedicated participants with NSSI and 84 healthy controls were included in the rsFC analysis. The NSSI group was subdivided into early-onset (onset before age 14) and late-onset (onset at or after age 14) groups.

Results: The early-onset NSSI group exhibited increased rsFC within the sensorimotor network (SMN), specifically between the postcentral and pre/postcentral gyri, relative to the late-onset NSSI group at the permutation level ($p_{\text{perm}} < 0.001$; matrix-wide $q_{\text{FDR}} = 0.211$). Nominal permutation-level associations between DERS scores and rsFC were also observed, but these did not reach the threshold for matrix-wide FDR correction and should be interpreted as exploratory.

Limitations: The exploratory, cross-sectional design and reliance on retrospective self-report of NSSI onset limit causal inference. The modest subgroup sample sizes and the exploratory nature of the findings may also affect the robustness of the results, highlighting the need for larger longitudinal studies.

Conclusions: These exploratory findings suggest that the age of NSSI onset may be associated with potentially distinct rsFC profiles. Pending longitudinal replication, onset age represents a key candidate variable for understanding the neurobiology of NSSI.

1. Introduction

Nonsuicidal self-injury (NSSI), defined as the deliberate, direct destruction of one's body tissue without suicidal intent (Nock, 2009), is a significant clinical concern. While it typically emerges during adolescence and young adulthood (Kiekens et al., 2023; Plener et al., 2018), there is an alarming trend of NSSI initiation at increasingly younger ages, often between 12 and 14 years (Muehlenkamp and Tillotson, 2024a; Muehlenkamp et al., 2019). Recent evidence indicates that NSSI is not only a symptom of other mental disorders but also a clinically meaningful condition primarily characterized by emotion dysregulation (Kuehn et al., 2022; Plener et al., 2018; Wolff et al., 2019; Andersson et al., 2022). This unique clinical significance is reflected in its

inclusion as a condition for further study in the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) (American Psychiatric Association, 2013). Despite this recognition, the neurophysiological mechanisms underlying NSSI remain poorly understood, particularly in relation to the age of onset and its role in phenotypic variations. Neuroimaging research that examines age-related differences and employs subgroup-specific analyses is necessary to clarify these mechanisms, especially since the age of onset is a known factor that significantly influences the diagnosis, assessment, and prognosis of many childhood-onset mental disorders (American Psychiatric Association, 2013; First et al., 2022).

Early-onset NSSI is associated with more severe self-injury behaviors, a poorer prognosis (Grandclerc et al., 2016; Muehlenkamp and

* Corresponding author.

E-mail address: j.hur@korea.ac.kr (J.-W. Hur).

¹ Contributed equally.

Brausch, 2019), and a heightened risk of transitioning to suicidal ideation and behaviors (Brager-Larsen et al., 2022; Turner et al., 2024). Moreover, younger individuals tend to engage in more frequent and versatile forms of NSSI (Ammerman et al., 2018). Although prior research has examined the relationship between the age of onset and NSSI severity, the pathological mechanisms underlying these differences remain unclear. Therefore, understanding the neural correlates of the age of onset of NSSI is crucial for identifying how this onset age relates to symptom severity and could help guide more effective interventions.

In this study, we investigated the differences in resting-state functional connectivity (rsFC) between early- and late-onset NSSI to understand the impact of the age of onset on neurophysiological profiles. RsFC, which reflects the synchronized neural activity among different brain regions during rest, has proven valuable for revealing nuanced neural correlates of mental health conditions and brain function changes (Canario et al., 2021). For example, rsFC studies have successfully detected distinctions between early- and late-onset subtypes in various mental disorders, including schizophrenia (Watsky et al., 2018), bipolar disorder (Cattarinussi et al., 2022), obsessive-compulsive disorder (Bruin et al., 2023), and major depressive disorder (Cai et al., 2021).

To explore the neurophysiological differences between early- and late-onset NSSI, we employed rsFC analysis using the Independent Component Analysis (ICA) (Calhoun et al., 2001), a multivariate data-driven approach. This study analyzed rsFC abnormalities in a large sample of 91 individuals with NSSI.

Emotion dysregulation is a central feature of non-suicidal self-injury (NSSI) (Wolff et al., 2019). This study investigates whether the age of onset is linked to distinct functional network architectures compared to healthy controls. Recent neuroimaging research highlights the importance of limbic and paralimbic circuits, such as the amygdala, insula, and anterior cingulate cortex, in NSSI (Niu et al., 2023). However, NSSI is primarily characterized by difficulty regulating emotion, not by a lack of emotional experience. Therefore, our framework emphasizes the cognitive-executive, self-referential, and sensorimotor circuits that modulate emotional states.

Drawing on the meta-analysis by Kohn et al. (2014) on cognitive emotion regulation and the emerging concept of sensory emotion regulation (Rodriguez & Kross, 2023), we propose a dual-network hypothesis for NSSI. First, we suggest a failure to integrate the default mode network (DMN, centered on the angular gyrus) with the central executive network (CEN, including the dorsolateral prefrontal cortex and middle temporal gyrus). This disconnection likely impairs top-down cognitive control over self-referential distress. Second, given that NSSI often involves the intentional use of physical pain to regulate affect, we propose that the sensorimotor network (SMN) may act as a transdiagnostic hub for impulsivity and psychomotor excitation (Baker et al., 2019; Kebets et al., 2019).

To contextualize these proposed network disruptions, it is important to distinguish the neuropathology of NSSI from other affective disorders. Increased DMN-CEN connectivity is common in major depressive disorder (Cheng et al., 2018; Zhang et al., 2022) and in depressive adolescents with NSSI (Ho et al., 2021), reflecting excessive cognitive coupling during rumination. However, we propose a different mechanism for NSSI. Recent meta-analyses indicate that rumination only modestly influences NSSI engagement (Cheung et al., 2024; Nagy et al., 2023). Therefore, we hypothesized that the age of NSSI onset may be associated with differential functional coupling patterns — potentially reflecting distinct trajectories of regulatory network development — which suggest altered executive and sensorimotor modulation. We systematically examined how the age of NSSI onset affects these networks and their relationship to emotion dysregulation. This approach may identify distinct neurophysiological markers for onset age and support more targeted diagnostic frameworks.

2. Methods

2.1. Participants

A total of 215 participants were recruited through online community advertisements. Initial screening was conducted via telephone to identify individuals who engaged in NSSI; eligible participants were required to confirm that they had engaged in NSSI more than five times in the previous year. Following telephone screening, participants visited the laboratory to complete the Structured Clinical Interview for DSM-5 (First et al., 2017) and the short forms of the Korean Wechsler Adult Intelligence Scale-IV using arithmetic and information subsets to estimate their intelligence (Choe et al., 2014).

All participants completed a series of self-report assessments, including the Inventory of Statements About Self-injury (ISAS) to evaluate the functionality and versatility of NSSI (Kim et al., 2019; Klonsky and Glenn, 2009), and the Difficulties in Emotion Regulation Scale to measure individual differences in emotion regulation challenges (Choi, 2007; Gratz and Roemer, 2004). Depression severity was measured using the Beck Depression Inventory-II (BDI-II) (Beck et al., 1996; Yu et al., 2011), anxiety using the Beck Anxiety Inventory (BAI) (Beck et al., 1988; Kwon, 1992; Yook & Kim, 1997), and alexithymia using the Toronto Alexithymia Scale-20 (TAS-20) (Bagby et al., 1994; Seo et al., 2009). BDI-II, BAI, and TAS-20 scores were used to exclude participants with significant clinical symptoms to ensure homogeneity within the HC group.

The exclusion criteria were as follows: [1] neurological diseases, [2] an estimated intelligence quotient <80, and [3] a family history of psychosis. Two participants with NSSI were excluded due to anatomical abnormalities or missing self-reported questionnaires. In addition, 18 eligible participants in the HC group were excluded from the neuroimaging analysis based on the following criteria: [1] history of NSSI, [2] presence of any mental disorder, [3] TAS-20 > 65 (mean + 1.5 standard deviation; SD), [4] DERS >104 (mean + 1 SD (Choi, 2007); a higher score than the mean of the NSSI group), [5] BDI-II > 13 (minimal range cut-off), and [6] BAI > 7 (minimal range cut-off). Another 20 participants were excluded because of the excessive head motion during the magnetic resonance imaging (MRI), resulting in a final sample of 91 unmedicated individuals with NSSI (78 females) and 84 HCs (65 females). The NSSI group was then divided into an early-onset group (age of onset <14 years, $n = 42$) and a late-onset group ($n = 49$), following previous literature (De Luca et al., 2023; Muehlenkamp et al., 2019). The Institutional Review Board approved the study of the university, and all participants provided written informed consent. The psychiatric diagnostic status and comorbidity profiles of the NSSI participants are summarized in Table 2.

2.2. Data acquisition

Resting-state fMRI (rs-fMRI) was performed using a 3 T Siemens Trio MRI scanner (Siemens Healthcare, Erlangen, Germany) with a 32-channel head coil. Scanning was conducted at the Seoul National University Brain Imaging Center and the Korea University Brain Imaging Center. Functional imaging data were collected using multi-band echo-planar imaging (EPI) over a 6-min and 44-s duration with the following parameters: TR/TE = 1500/30 ms, flip angle = 90°, field of view = 256 mm, voxel size = 2.3 × 2.3 × 2.3 mm, 64 slices in an interleaved sequence, phase encoding (PE) in the anterior-posterior direction, GRAPPA factor = 2, and multi-band acceleration factor = 4.

2.3. Data preprocessing

T1-weighted and rs-fMRI data were preprocessed using the standard preprocessing pipeline of the CONN toolbox v.19 (Whitfield-Gabrieli & Nieto-Castanon, 2012) implemented in MATLAB system (version R2022a, MathWorks, Inc., Natick, MA, USA). Slice-timing correction

was applied, followed by realignment for motion correction. Motion-related variance was accounted for using spatial parameters from the motion correction process. EPI images were then normalized to the Montreal Neurological Institute template and smoothed using a 6-mm full-width at half maximum Gaussian kernel. Twenty participants with a mean frame-wise displacement exceeding 0.2 were excluded from further analyses (Power et al., 2012). No significant group differences were observed in the frame-wise displacement values.

2.4. Group independent component analysis (GICA)

The GICA was conducted using the Group ICA of the fMRI Toolbox (GIFT v3.0c, <http://mialab.mrn.org/software/gift/>). Higher-order ICA has been shown to enable finer-grained parcellation of resting-state networks, reflecting the hierarchical organization of brain functional architecture (Allen et al., 2011). In particular, prior work has demonstrated that sensitivity to between-group differences in functional connectivity varies as a function of model order, with peak sensitivity observed around 70 components, suggesting this range as a practical balance between functional specificity and over-fragmentation (Abou Elseoud et al., 2011). Based on this evidence, 70 ICs were adopted for the current analysis (Kiviniemi et al., 2009; Abou Elseoud et al., 2011;

Allen et al., 2011).

Preprocessed rs-fMRI data were normalized to Z-scores. Dimensionality reduction was performed using a two-stage PCA approach. First, for each subject, PCA was applied to reduce the data to 105 principal components, thereby decreasing dimensionality and computational burden while preserving subject-level variance. The reduced data were then temporally concatenated across subjects. In the second stage, a group-level PCA using an expectation-maximization algorithm reduced the dimensionality to 70 principal components. These data were subsequently entered into ICA decomposition with a predefined model order of 70 components. The Infomax ICA algorithm was repeated 20 times using the ICASSO toolbox (Himberg et al., 2004) to extract the most stable component clusters at the group level; all components had a stability index above 0.9. The GICA back-reconstruction algorithm (Calhoun et al., 2001) was used to estimate subject-specific spatial maps and time courses, which were then transformed to Z-scores.

We classified the resulting ICs as signals or noise (e.g., physiological artifacts or head movements). Two trained authors (JK, HM) independently evaluated each IC region as either a resting-state component or noise using guidelines from previous work (Allen et al., 2011; Griffanti et al., 2017) considering the following parameters: [1] the component was primarily located in the grey matter; [2] it did not overlap with

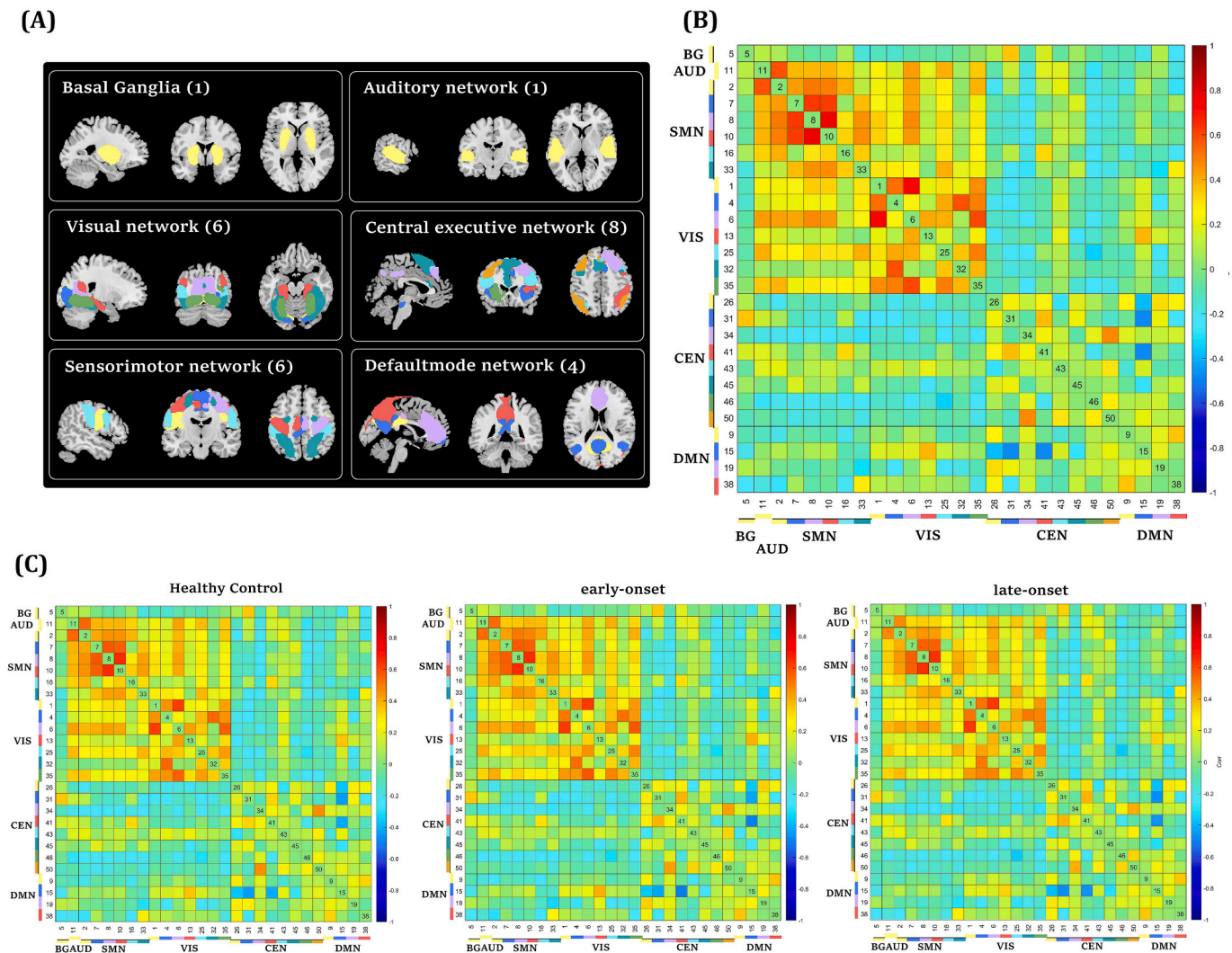


Fig. 1. The resting-state functional network and connectivity pattern. (A) Six functional networks (BG, basal ganglia; AUD, auditory network; VIS, visual network; SMN, sensorimotor network; DMN, default mode network; and CEN, cognitive executive network) identified from the 27 independent components (ICs). (B) Correlation matrix of all participants displaying static functional network connectivity for IC pairs. (C) Connectivity matrices for each group: healthy control, early-onset, and late-onset NSSI.

vascular, ventricular, or susceptibility artifacts; and [3] its time course was dominated by low-frequency signals (with a low power ratio between frequencies below 0.1 Hz and those within 0.15–0.25 Hz). Ultimately, 27 resting-state independent components were identified and categorized into six distinct functional networks (Smith et al., 2009): basal ganglia (BG), auditory (AUD), visual (VIS), sensorimotor (SMN), cognitive executive (CEN), and default mode (DMN), as illustrated in Fig. 1A.

2.5. rsFC Analysis

The rsFC matrices were computed as pairwise Pearson correlations between the IC time series. These correlations were converted to Z-scores using Fisher's Z-transformation. Prior to group-level analyses, head motion effects were addressed at the individual level by regressing out the six rigid-body motion parameters from each component time course using a general linear model within the MANCOVAN toolbox. To assess and control for site-related variance, ComBat harmonization (Johnson et al., 2007) was applied to the Fisher Z-transformed FNC matrices using the ComBatHarmonization MATLAB toolbox (Fortin (https://github.com/Jfortin1/ComBatHarmonization; Fortin et al., 2017)). Following harmonization, no FNC features retained significant site differences (0 of 351), with the mean absolute site difference reduced by 59.7%. The harmonized FNC matrices were subsequently analyzed using Analysis of Covariance (ANCOVA) with the Multivariate Analysis of Covariance (MANCOVAN) toolbox in GIFT (Allen et al., 2011). Analyses included group and DERS interaction factors, with nuisance covariates for sex, age, BDI-II, BAI, handedness, and mean framewise displacement (FD). To assess the statistical significance of group comparison and interaction effects, permutation-based testing was applied. For group comparison contrasts, group labels were permuted 5000 times to construct empirical null distributions. For Group $\times$ DERS interaction contrasts, the Freedman-Lane permutation method (Freedman and Lane, 1983) was employed, in which residuals from the reduced null model (excluding interaction terms) were permuted and added back to the null model fitted values to construct empirical null distributions, as recommended for interaction testing in regression models (Bůžková et al., 2011). Benjamini-Hochberg FDR correction was applied across all 351 permutation-derived p -values to account for multiple comparisons. Both the edge-wise uncorrected permutation p -values (p_{perm}) and the FDR-corrected q -values (q_{FDR}) are reported in Table 3.

2.6. Statistical analysis

Demographic and clinical data were analyzed using the IBM SPSS Statistics 27 (IBM Corporation, Armonk, NY, USA). Two-sample t -tests were used to compare age, NSSI functionality, NSSI versatility, DERS, TAS-20, BDI-II, and BAI scores between the early- and late-onset groups. ANOVA was used to assess group differences in the clinical data, excluding NSSI-related scales. Sex and handedness distributions were compared using χ^2 tests. Statistical significance was set at $p < 0.05$.

3. Results

3.1. Demographic and clinical characteristics

The early- and late-onset NSSI groups did not differ significantly on demographic variables, including age ($t = -0.46$, $p = 0.650$, Cohen's $d = -0.09$, 95% CI $[-0.49, 0.31]$), sex ($\chi^2 = 0.00$, $p = 1.000$), or handedness ($\chi^2 = 0.78$, $p = 0.376$), nor on NSSI-specific behavioral measures, including NSSI versatility ($t = 1.13$, $p = 0.260$, Cohen's $d = 0.24$, 95% CI $[-0.18, 0.66]$). Critically, the two NSSI subgroups were also comparable across all primary clinical measures (DERS scores, $t = -0.54$, $p = 0.590$, Cohen's $d = -0.11$; BDI-II scores, $t = -1.70$, $p = 0.093$, Cohen's $d = -0.36$; BAI scores, $t = -1.43$, $p = 0.157$, Cohen's $d = -0.30$).

Alexithymia (measured by TAS-20 scores) did not show group differences ($t = -1.44$, $p = 0.153$, Cohen's $d = -0.31$).

When each NSSI subgroup was compared with healthy controls, both the early- and late-onset NSSI groups showed significantly higher BDI-II and BAI scores than HCs (all $p < 0.05$; see Table 1). Similarly, both NSSI groups exhibited significantly elevated DERS scores compared to HCs. Notably, there were no significant differences between early- and late-onset NSSI groups on these clinical measures. The comorbidity profiles of both NSSI subgroups, including current diagnoses and clinical features, are detailed in Table 2.

3.2. Neuroimaging results

3.2.1. Identified rsFC networks

Fig. 1A shows the integrated spatial maps of the 27 ICs identified using group ICA. These ICs were categorized into six distinct networks: the BG (IC 5), AUD (IC 11), VIS (ICs 1, 4, 6, 13, 25, 32, and 35), SMN (ICs 2, 7, 8, 10, 16, and 33), CEN (frontoparietal network ICs 26, 34, 41, and 46; salience network ICs 31 and 45; attention network ICs 43 and 50), and DMN (ICs 9, 15, 19, and 38). Fig. 1B shows the rsFC network computed across all ICs and averaged among the participants. Detailed information on these ICs and their spatial distributions is provided in eTable 1 and eFigure 1 in Supplement 1.

3.2.2. rsFC Patterns Discriminating Early-Onset NSSI from HC and Late-Onset NSSI

No significant differences in rsFC were observed between the early-onset NSSI and HC groups across any of the networks examined. However, comparative analysis between the early- and late-onset NSSI groups revealed distinct rsFC patterns in the SMN. Specifically, the early-onset group displayed increased rsFC between IC08 (postcentral gyrus) and IC16 (pre/postcentral gyrus) compared to the late-onset group at the edge-wise permutation level ($p_{\text{perm}} < 0.001$), which corresponds to a matrix-wide FDR-corrected q -value of 0.211 across the 351 IC pairs. The paired ICs are shown in Fig. 2.

3.2.3. rsFC patterns differentiating late-onset NSSI from HC

No significant differences in rsFC were observed between the late-onset NSSI and HC groups across any of the networks examined.

3.2.4. rsFC and its relevance to emotion dysregulation

The MANCOVAN interaction revealed a significant group difference in the DERS–IC09–IC43 relationship ($p_{\text{perm}} < 0.001$): the late-onset group showed a positive association ($r = 0.28$), whereas the early-onset group showed the opposite pattern ($r = -0.22$), though neither within-group correlation survived FDR correction independently. In the early-onset NSSI group, DERS scores positively correlated with rsFC strength in several network components, including IC08 (postcentral gyrus) and IC16 (pre/postcentral gyrus; IC08–IC16), as well as IC06 (lingual gyrus) and IC16 (IC06–IC16) (Fig. 3A, B). The opposing directional trends between onset groups for IC09–IC43 are illustrated in Fig. 3C. Importantly, onset-specific associations were observed between rsFC and emotion dysregulation across multiple neural networks, including the visual, sensorimotor, default mode, and central executive networks. Overall, early- and late-onset NSSI groups demonstrated differential rsFC patterns, particularly within emotion regulation and sensorimotor systems (See Fig. 4). At the permutation level, nominal associations were observed between DERS scores and rsFC across several network components; however, none survived FDR correction across 351 IC pairs (all $q_{\text{FDR}} > 0.05$) and all should be regarded as exploratory. A comprehensive summary of IC pairs, network labels, directional effects, and correction statistics is provided in Table 3.

Table 1
Demographic and Clinical Characteristics of Participants.

Variable	Mean (SD)			Group Comparison		Early vs. Late		
	Early-Onset NSSI (n = 42)	Late-Onset NSSI (n = 49)	Healthy Control (n = 84)	F / χ^2	p	t / χ^2	p	Cohen's d [95% CI]
Demographic Variables								
Age, y	21.67 (2.51)	21.92 (2.76)	22.30 (2.55)	F = 0.90	0.409	t = -0.46	0.650	-0.09 [-0.49, 0.31]
Male/Female, n	6/36	8/41	20/64	$\chi^2 = 2.04$	0.360	$\chi^2 = 0.00$	1.000	—
Handedness, n (right/left)	40/2	43/6	75/9	$\chi^2 = 1.63$	0.443	$\chi^2 = 0.78$	0.376	—
NSSI-Specific Measures (NSSI participants only)								
NSSI functions ^a , intrapersonal	0.90 (0.49)	0.84 (0.38)	—	—	—	t = 0.67	0.505	—
interpersonal	0.32 (0.23)	0.30 (0.27)	—	—	—	t = 0.31	0.513	—
NSSI versatility	4.76 (2.53)	4.18 (2.30)	—	—	—	t = 1.13	0.260	0.24 [-0.18, 0.66]
Clinical Measures								
DERS	100.55 (24.16)	103.51 (28.19)	57.89 (15.42)	F = 90.32	< 0.001	t = -0.54	0.590	-0.11 [-0.53, 0.30]
TAS-20	53.21 (12.55)	56.96 (10.18)	37.46 (7.59)	F = 82.48	< 0.001	t = -1.44	0.153	-0.31 [-0.73, 0.10]
BDI-II	20.48 (11.09)	24.22 (9.74)	4.49 (3.70)	F = 116.12	< 0.001	t = -1.70	0.093	-0.36 [-0.82, 0.05]
BAI	12.90 (11.89)	16.24 (10.16)	1.54 (1.72)	F = 61.18	< 0.001	t = -1.43	0.157	-0.30 [-0.78, 0.11]

Note. BDI-II = Beck Depression Inventory-II; DERS = Difficulties in Emotion Regulation Scale; TAS-20 = Toronto Alexithymia Scale-20; BAI = Beck Anxiety Inventory. Cohen's d and 95% CI pertain to the Early-onset vs. Late-onset comparison. DASH (—) indicates no Cohen's d reported for categorical variables.

^a NSSI functions measured by ISAS. Versatility and function of NSSI were assessed exclusively in NSSI participants; group comparison test not applicable (—).

Table 2
Psychiatric Comorbidity Profile of NSSI Participants by Onset Group.

Diagnosis / Feature	Early-Onset NSSI (n = 42)		Late-Onset NSSI (n = 49)		Total NSSI (n = 91)	
	n	(%)	n	(%)	n	(%)
No comorbidity	12	(28.6)	12	(24.5)	24	(26.4)
Current diagnoses (Comorbidities)						
Depressive disorders						
MDD (current episode)	4	(9.5)	9	(18.4)	13	(14.3)
PDD	6	(14.3)	15	(30.6)	21	(23.1)
Bipolar and related disorders						
Bipolar II Disorder	0	(0.0)	2	(4.1)	2	(2.2)
Anxiety disorders						
GAD	4	(9.5)	9	(18.4)	13	(14.3)
Social Anxiety Disorder	1	(2.4)	3	(6.1)	4	(4.4)
Panic Disorder	2	(4.8)	1	(2.0)	3	(3.3)
Agoraphobia	1	(2.4)	1	(2.0)	2	(2.2)
Specific Phobia	1	(2.4)	0	(0.0)	1	(1.1)
Other Specified Anxiety Disorder	0	(0.0)	1	(2.0)	1	(1.1)
Trauma- and stress-related						
PTSD	2	(4.8)	4	(8.2)	6	(6.6)
Obsessive-compulsive disorders						
OCD	1	(2.4)	2	(4.1)	3	(3.3)
Neurodevelopmental disorders						
ADHD	2	(4.8)	2	(4.1)	4	(4.4)
Feeding and eating disorders						
Eating Disorder	2	(4.8)	0	(0.0)	2	(2.2)
Substance-related disorders						
Alcohol/Substance Use Disorder	9	(21.4)	11	(22.4)	20	(22.0)
History of mood disorder						
History of MDD	15	(35.7)	20	(40.8)	35	(38.5)
History of Bipolar Disorder	0	(0.0)	1	(2.0)	1	(1.1)
Additional clinical features						
Insomnia	2	(4.8)	1	(2.0)	3	(3.3)
Suicidal ideation / attempt	3	(7.1)	4	(8.2)	7	(7.7)

Note. Values represent n (%). Percentages are calculated within each onset group. Participants may carry more than one diagnosis; percentages may not sum to 100%. Diagnoses established via SCID-5. PDD = Persistent Depressive Disorder; MDD = Major Depressive Disorder; GAD = Generalized Anxiety Disorder; PTSD = Post-Traumatic Stress Disorder; OCD = Obsessive-Compulsive Disorder; ADHD = Attention-Deficit/Hyperactivity Disorder.

4. Discussion

This study is the first to identify distinct rsFC patterns between the early- and late-onset NSSI groups and their association with emotion dysregulation, a core feature of NSSI. Our key findings demonstrate that the early-onset NSSI group exhibited increased rsFC within the SMN, specifically between the postcentral and pre/postcentral gyri, compared to the late-onset group. Furthermore, onset-specific associations between rsFC and emotion dysregulation revealed differential regulatory profiles: the early-onset group showed nominal SMN-centered associations at the permutation level that did not reach the threshold for matrix-wide FDR correction and should be interpreted as exploratory. The MANCOVAN interaction identified a group-differential DERS-DMN-VLPFC relationship that similarly warrants further investigation in larger samples. Specifically, the late-onset group showed a trend toward increased DMN-VLPFC coupling under conditions of heightened emotional distress. These preliminary findings offer hypothesis-generating insight into the neurophysiological correlates of NSSI onset age.

4.1. Neural patterns of rsFC in early-onset NSSI

A key finding is that the early-onset group demonstrated increased rsFC within the SMN, particularly between the postcentral and precentral gyri, compared to the late-onset group. A defining clinical and perceptual feature of NSSI is the deliberate instrumentalization of physical pain to achieve affective relief. These observations broadly align with the emerging framework of *sensory emotion regulation* (Rodríguez & Kross, 2023), providing tentative support for the idea that somatosensory inputs may assume a primary regulatory role when higher-order regulatory demands are not met through prefrontal circuits. In essence, physical sensation acts as a proxy for the cognitive modulation of affect. This account is further supported by prior research identifying the SMN as a transdiagnostic hub for impulsivity, psychomotor excitation, and general psychopathology (Baker et al., 2019; Doucet et al., 2017; Kebets et al., 2019; Martino et al., 2016).

In addition, the persistence of SMN hyperconnectivity into young adulthood is notable. For individuals who initiated NSSI before age 14, this sensory-based regulatory strategy may be associated with early developmental experiences, though prospective studies are needed to

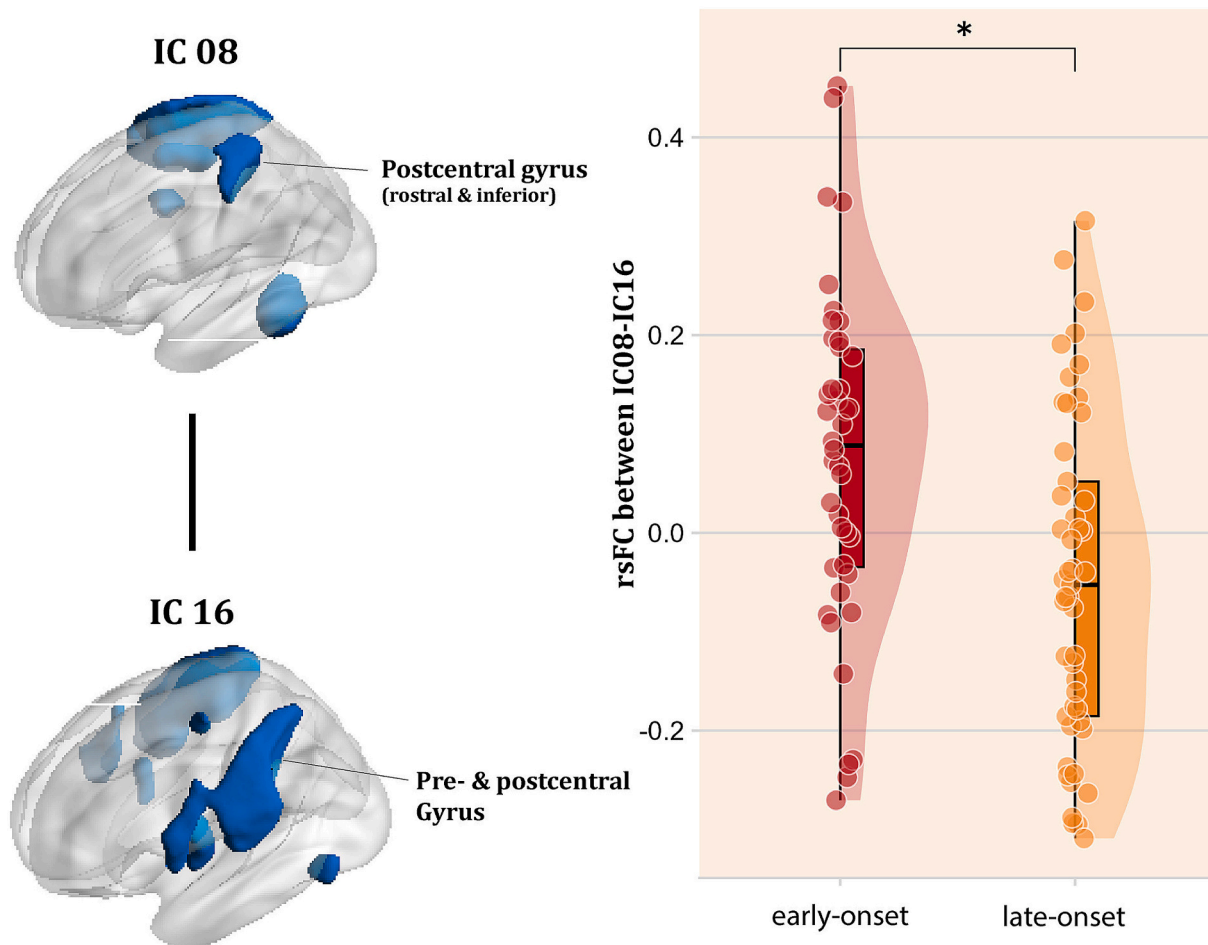


Fig. 2. Main group differences in resting-state functional connectivity (rsFC). rsFC alterations between the early- and late-onset NSSI groups. The early-onset group showed increased rsFC between IC08 (postcentral gyrus) and IC16 (precentral/postcentral gyrus) compared to the late-onset group.

confirm this consolidation (Ammerman et al., 2018).

4.2. Emotion dysregulation and rsFC in early-onset NSSI

The relationship between rsFC and emotion dysregulation was notably modulated by the age of NSSI onset, revealing differential neural patterns, especially in the early-onset group. Greater emotion dysregulation in this group correlated with increased rsFC within the SMN, particularly in the postcentral gyrus, as well as with increased connectivity between the lingual gyrus (VIS) and the pre-/postcentral gyrus (SMN). This exploratory pattern may suggest that, as emotional dysregulation intensifies, early-onset individuals could preferentially recruit sensorimotor and perceptual circuits for affect regulation, a possibility compatible with the sensory emotion regulation framework (Rodriguez & Kross, 2023). Rather than engaging higher-order prefrontal circuits to modulate distress, regulatory effort appears to be channeled through somatosensory processing — a pattern that may reflect the behavioral reliance on physical sensation characteristic of NSSI (Kropf et al., 2018).

We propose that this dysregulated sensory-executive coupling may reflect a preliminary neurophysiological correlate of early-onset history, potentially linked with somatosensory-based regulation in adulthood. Taken together, the dysconnectivity patterns identified in this study may imply a candidate adult neural correlate of an early-onset history, one characterized by a somatosensory-dependent regulatory mode, in which physical sensation substitutes for normative higher-order affect regulation. Whether this represents a pre-existing neurobiological vulnerability or neurobiological scarring from repeated self-injury during sensitive developmental windows remains to be determined. These

distinct rsFC patterns suggest that NSSI onset age may be associated with distinct neural circuits involved in emotion processing. These preliminary findings warrant further investigation with longitudinal studies to determine whether onset-related neural patterns might inform future research into more tailored approaches to emotion dysregulation.

4.3. SMN connectivity in early-onset NSSI

At the permutation level, the early-onset group showed greater rsFC within the SMN (IC08–IC16; $p_{\text{perm}} < 0.001$), though this did not reach the threshold for matrix-wide FDR correction across all 351 IC pairs ($q_{\text{FDR}} = 0.211$). This exploratory connectivity difference is consistent with the emerging framework of sensory emotion regulation (Rodriguez & Kross, 2023) and prior work identifying the SMN as a transdiagnostic hub for impulsivity and psychomotor excitation (Baker et al., 2019; Kebets et al., 2019). These observations are hypothesis-generating and require replication in larger, adequately powered samples before any definitive inference can be drawn.

4.4. Clinical-neural dissociation in late-onset NSSI

A theoretically significant finding is the absence of stable, resting-state rsFC differences in the late-onset NSSI group, despite their severe clinical profile. We propose that the neurobiological profile of late-onset NSSI may be fundamentally state-dependent rather than trait-stable. The MANCOVA Group $\times$ DERS interaction revealed differential DMN–VLPFC coupling between onset groups ($p_{\text{perm}} < 0.001$): within the late-onset group, higher emotion dysregulation was associated with a

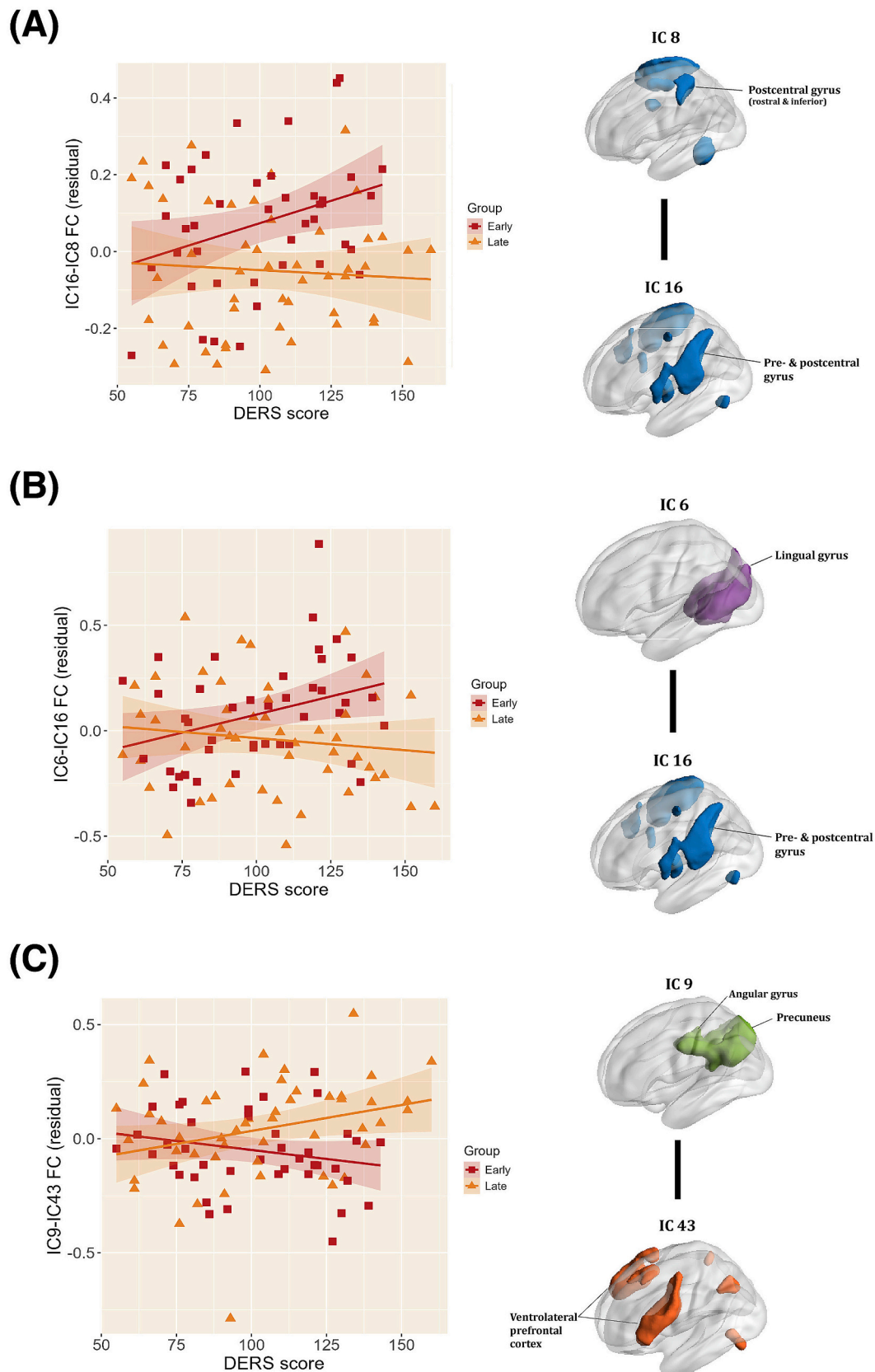


Fig. 3. Correlation plots of rsFC alterations between the early- and late-onset NSSI groups with the interaction effect of DERS scores. (A, B): Positive correlations between DERS scores and rsFC strength in the early-onset NSSI group across various network components. These relationships were not observed in the late-onset group. (C) A positive correlation between DERS scores and rsFC between the IC09 (precuneus/angular gyrus) and IC43 (VLPFC/inferior frontal gyrus) in the late-onset NSSI group, in contrast to a negative correlation in the early-onset group, highlighting distinct emotion dysregulation patterns between the two groups. Note: These associations did not survive FDR correction across 351 IC pairs (all $q_{FDR} > 0.05$) and should be interpreted as exploratory.

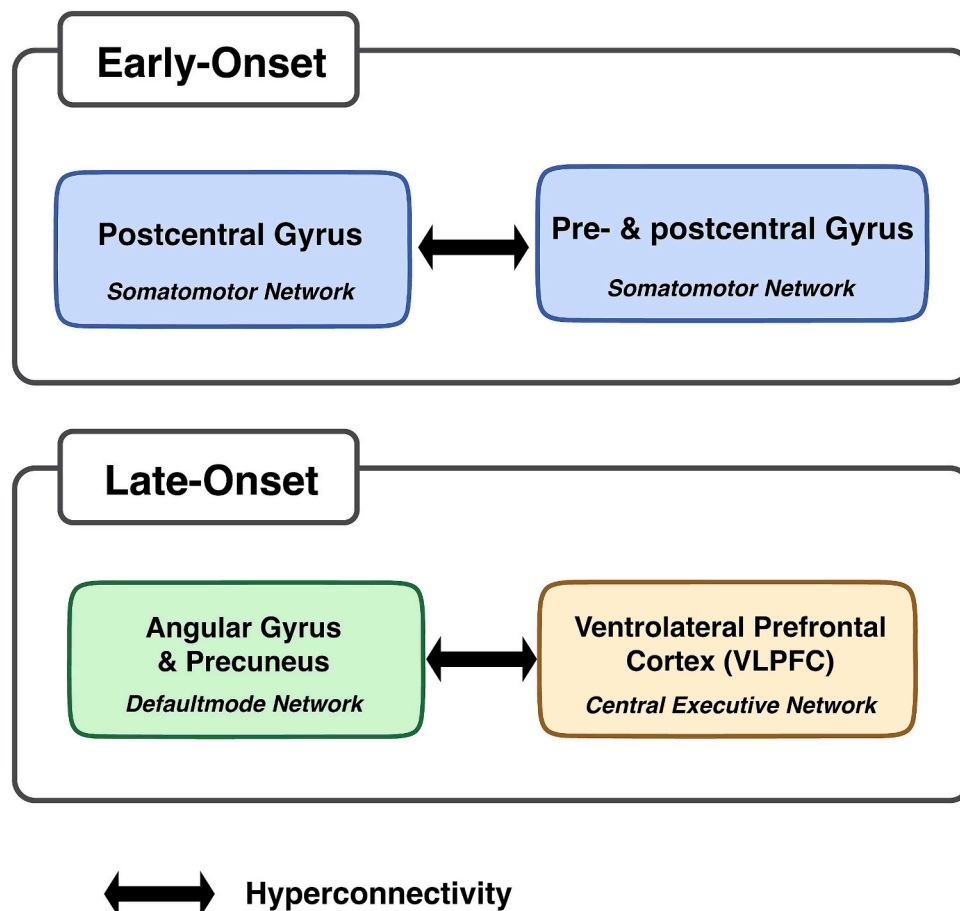


Fig. 4. Schematic illustration of onset-specific functional network profiles. The top panel depicts the early-onset NSSI group, characterized by hyperconnectivity within the sensorimotor network (SMN; IC08–IC16: postcentral gyrus – pre-/postcentral gyrus), which was significantly elevated relative to the late-onset group. And the bottom panel depicts the hyperconnectivity between the default mode network and central executive network in late-onset NSSI group. Edges are non-directional, reflecting the correlation-based nature of resting-state functional connectivity.

Table 3
Summary of Significant Independent Component Pairs, Network Labels, Effect Sizes, and Correction Status.

Direction	IC Pair	Networks	IC (Region A)	IC (Region B)	DERS r (95% CI)	Statistics
Section 1. Group Contrast (Early-onset NSSI vs. Late-onset NSSI)						
Hyperconnectivity (Early > Late)	IC08–IC16	SMN–SMN	IC08: Postcentral gyrus	IC16: Pre-/postcentral gyrus	–	$t = 3.37$, $d = 0.75$, $\eta^2 = 0.062$ ($p_{\text{perm}} = 0.0006$ / $q\text{FDR} = 0.211$)
Section 2. Group (Early-onset NSSI vs. Late-onset NSSI) $\times$ DERS Interaction Effect (MANCOVA)						
Early + / Late –	IC08–IC16	SMN–SMN	IC08: Postcentral gyrus	IC16: Pre-/postcentral gyrus	Early-onset: $r = 0.330^{\dagger}$ [0.029, 0.576] Late-onset: $r = -0.065$ [–0.340, 0.220]	$t = 3.61$, $\eta^2 = 0.071$ ($p_{\text{perm}} = 0.0002$ / $q\text{FDR} = 0.070$)
Early + / Late –	IC06–IC16	VIS–SMN	IC06: Lingual gyrus	IC16: Pre-/postcentral gyrus	Early-onset: $r = 0.331^{\dagger}$ [0.031, 0.577] Late-onset: $r = -0.130$ [–0.397, 0.157]	$t = 3.37$, $\eta^2 = 0.062$ ($p_{\text{perm}} = 0.0006$ / $q\text{FDR} = 0.094$)
Early – / Late +	IC09–IC43	DMN–CEN (VLPFC)	IC09: Precuneus / Angular gyrus	IC43: VLPFC / Inferior frontal gyrus	Early-onset: $r = -0.217$ [–0.489, 0.093] Late-onset: $r = 0.284^{\dagger}$ [0.003, 0.524]	$t = -3.29$, $\eta^2 = 0.059$ ($p_{\text{perm}} = 0.0008$ / $q\text{FDR} = 0.094$)

Note. IC = independent component; SMN = sensorimotor network; VIS = visual network; DMN = default mode network; CEN = central executive network; VLPFC = ventrolateral prefrontal cortex; DERS = Difficulties in Emotion Regulation Scale; rsFC = resting-state functional connectivity; FDR = false discovery rate; p_{perm} = permutation-derived p -value; $q\text{FDR}$ = FDR-corrected q -value; n.s. = not significant.

d = Cohen's d ; η^2 = partial eta-squared; DERS r = within-group Pearson correlation between DERS and rsFC for pairs showing significant Group $\times$ DERS interaction effect (exploratory; † 95% CI excludes zero).

All permutation tests used 5000 permutations. FDR correction (Benjamini-Hochberg) was applied across 351 tests.

trend toward increased coupling between the precuneus/angular gyrus (IC09) and the VLPFC/inferior frontal gyrus (IC43; $r = 0.28$, $p_{\text{perm}} <$

0.001), whereas the early-onset group showed the opposite directional trend ($r = -0.22$). These opposing patterns are consistent with the view that late-onset neuropathology may emerge primarily under conditions of emotional challenge. The VLPFC is critically implicated in voluntary emotion regulation through cognitive reappraisal (Kohn et al., 2014; Ochsner et al., 2012; Seo et al., 2014; Khodadadifar et al., 2022), while the precuneus and angular gyrus anchor self-referential and ruminative processes within the DMN (Kim et al., 2023). The group-level interaction is compatible with the possibility that, under emotional load, late-onset individuals engage — yet may fail to efficiently deploy — a prefrontal regulatory circuit targeting self-referential distress. However, this interpretation remains exploratory and requires replication in larger samples. Furthermore, late-onset NSSI may be etiologically more heterogeneous, precipitated by diverse acute psychosocial stressors rather than a uniform early developmental disruption (Kaess et al., 2021; Li et al., 2026). This heterogeneity reduces the likelihood of detecting a consistent group-level neural signal at rest and underscores the need for task-based or ecological momentary fMRI paradigms that can capture state-dependent regulatory dynamics in this subgroup. Finally, in addition to these theoretical considerations, methodological factors such as modest sample size and the categorical age-14 cutoff may have also contributed to the absence of stable baseline differences.

5. Limitations

This study has several limitations. First, since this is an initial exploration of onset-age-defined neurophysiological subgroups in NSSI, we used a relatively liberal statistical threshold. As a result, our findings need to be replicated in larger, better-powered samples with improved control for comorbidities. Second, our cross-sectional design and reliance on retrospective self-reports introduce significant interpretive constraints regarding developmental timing. Using retrospective recall to determine the age of onset raises the risk of misclassification, especially for cases near the age-14 threshold. There is also a large time gap between the neural assessment in young adulthood (ages 19–29) and the reported onset period (before age 14). Because of this gap, we cannot distinguish the effects of early onset from closely related factors such as illness duration or cumulative severity of self-harm over time. Thus, our findings should be viewed specifically as the adult neural signature of an early-onset history rather than as a direct reflection of the brain at the time of onset. Third, our exclusive focus on static between-network connectivity leaves both intra-network organization and temporal dynamics unexamined. Given the highly episodic nature of emotion dysregulation in NSSI, static metrics may not fully capture the transient network reconfigurations associated with acute distress. Future research employing dynamic functional connectivity (e.g., Zhang et al., 2024) and intra-network coherence analyses (e.g., Zhang et al., 2023) will be critical to determine whether the dysconnectivity observed here reflects persistently aberrant coupling or a reduced capacity to flexibly transition between necessary regulatory states. Finally, our strict screening criteria for healthy controls, including the exclusion of individuals with subclinical symptoms, resulted in a highly purified, low-symptom reference sample that does not represent the broader nonclinical population. Although this approach reduced baseline variance, it may have inflated effect sizes in broad NSSI-versus-control comparisons and limited generalizability to real-world contexts, where subclinical emotional disturbances are common. Readers should consider these constraints when interpreting broad group differences. Importantly, this methodological limitation does not affect the validity of the study's main aim, as direct comparisons between NSSI subgroups utilize samples from the same clinical population. However, the nominal associations identified between emotion dysregulation and functional connectivity should be interpreted as preliminary, as they did not reach the threshold for matrix-wide FDR correction. These patterns may partly reflect shared variance with generalized affective distress, suggesting that future studies should formally control for comorbid symptoms during the

enrollment phase.

6. Conclusions

Our findings suggest that the age of NSSI onset is associated with potentially distinct rsFC profiles that persist into young adulthood. Specifically, a history of early-onset NSSI was associated with greater SMN connectivity relative to the late-onset group. While these preliminary observations and their associations with emotion dysregulation represent hypothesis-generating insights, they provide a candidate neural framework for understanding the neurophysiological correlates of NSSI onset age. Pending prospective replication in larger longitudinal cohorts, onset age represents a key candidate variable for future investigations into the distinct neural circuits underlying emotion regulation in NSSI. These findings lay a necessary foundation for future research into onset-specific neural mechanisms of self-injurious behaviors.

CRedit authorship contribution statement

Hyeri Moon: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Jin-hee Kim:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Gyumyoung Kim:** Writing – review & editing, Validation. **Hackjin Kim:** Writing – review & editing, Validation. **Taekwan Kim:** Writing – review & editing, Validation. **Ji-Won Hur:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

All authors report no biomedical financial interests or potential conflicts of interest.

Acknowledgments

This research was supported by the Brain Research Program of the National Research Foundation (NRF) (grant number: NRF-2022R1A2C2011467) funded by the Korea government (MSIT) and grant of the Mental Health related Social Problem Solving Project (grant number: RS-2024-00406503) funded by the Ministry of Health & Welfare, Republic of Korea.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jad.2026.122238>.

References

- Abou Elseoud, A., Littow, H., Remes, J., Starck, T., Nikkinen, J., Nissilä, J., Timonen, M., Tervonen, O., Kiviniemi, V., 2011. Group-ICA model order highlights patterns of functional brain connectivity. *Frontiers in Systems Neuroscience*. 5, 37. <https://doi.org/10.3389/fnsys.2011.00037>.
- Allen, E.A., Erhardt, E.B., Damaraju, E., Gruner, W., Segall, J.M., Silva, R.F., Havlicek, M., Rachakonda, S., Fries, J., Kalyanam, R., 2011. A baseline for the multivariate comparison of resting-state networks. *Frontiers in Systems Neuroscience* 5, 2. <https://doi.org/10.3389/fnsys.2011.00002>.
- American Psychiatric Association, 2013. *Diagnostic and statistical manual of mental disorders, DSM-5 (Vol. 5)*. American Psychiatric Association, Washington, DC. <https://doi.org/10.1176/appi.books.9780890425596>.
- Ammerman, B.A., Jacobucci, R., Kleiman, E.M., Uyeji, L.L., McCloskey, M.S., 2018. The relationship between nonsuicidal self-injury age of onset and severity of self-harm. *Suicide and Life-Threatening Behavior* 48 (1), 31–37. <https://doi.org/10.1111/sltb.12330>.
- Andersson, H., Aspegqvist, E., Dahlström, Ö., Svedin, C.G., Jonsson, L.S., Landberg, Å., Zetterqvist, M., 2022. Emotional dysregulation and trauma symptoms mediate the relationship between childhood abuse and nonsuicidal self-injury in adolescents. *Frontiers in Psychiatry* 13, 897081. <https://doi.org/10.3389/fpsy.2022.897081>.

- Bagby, R.M., Parker, J.D., Taylor, G.J., 1994. The twenty-item Toronto Alexithymia Scale—I. Item selection and cross-validation of the factor structure. *Journal of Psychosomatic Research* 38 (1), 23–32. [https://doi.org/10.1016/0022-3999\(94\)90005-1](https://doi.org/10.1016/0022-3999(94)90005-1).
- Baker, J.T., Dillon, D.G., Patrick, L.M., Roffman, J.L., Brady Jr., R.O., Pizzagalli, D.A., Öngür, D., Holmes, A.J., 2019. Functional connectomics of affective and psychotic pathology. *Proceedings of the National Academy of Sciences* 116 (18), 9050–9059. <https://doi.org/10.1073/pnas.1820780116>.
- Beck, A.T., Epstein, N., Brown, G., Steer, R.A., 1988. An inventory for measuring clinical anxiety: psychometric properties. *Journal of consulting and clinical psychology* 56 (6), 893. <https://doi.org/10.1037/0022-006X.56.6.893>.
- Beck, A.T., Steer, R.A., Brown, G.K., 1996. *Beck depression inventory*.
- Brager-Larsen, A., Zeiner, P., Klungsoyr, O., Mehlum, L., 2022. Is age of self-harm onset associated with increased frequency of non-suicidal self-injury and suicide attempts in adolescent outpatients? *BMC Psychiatry* 22 (1), 58. <https://doi.org/10.1186/s12888-022-03712-w>.
- Bruin, W.B., Abe, Y., Alonso, P., Anticevic, A., Backhausen, L.L., Balachander, S., Bargallo, N., Batistuzzo, M.C., Benedetti, F., Bertolin Triquell, S., 2023. The functional connectome in obsessive-compulsive disorder: resting-state mega-analysis and machine learning classification for the ENIGMA-OCD consortium. *Molecular Psychiatry* 28 (10), 4307–4319. <https://doi.org/10.1038/s41380-023-02077-0>.
- Bůžková, P., Lumley, T., Rice, K., 2011. Permutation and parametric bootstrap tests for gene-gene and gene-environment interactions. *Annals of Human Genetics* 75, 36–45. <https://doi.org/10.1111/j.1469-1809.2010.00572.x>.
- Cai, Y., Elsayed, N.M., Barch, D.M., 2021. Contributions from resting state functional connectivity and familial risk to early adolescent-onset MDD: Results from the Adolescent Brain Cognitive Development study. *Journal of Affective Disorders* 287, 229–239. <https://doi.org/10.1016/j.jad.2021.03.031>.
- Calhoun, V.D., Adali, T., Pearlson, G.D., Pekar, J.J., 2001. A method for making group inferences from functional MRI data using independent component analysis. *Human Brain Mapping* 14 (3), 140–151. <https://doi.org/10.1002/hbm.1048>.
- Canario, E., Chen, D., Biswal, B., 2021. A review of resting-state fMRI and its use to examine psychiatric disorders. *Psychoradiology* 1 (1), 42–53. <https://doi.org/10.1093/psyrad/kkab003>.
- Cattarinussi, G., Bellani, M., Maggioni, E., Sambataro, F., Brambilla, P., Delvecchio, G., 2022. Resting-state functional connectivity and spontaneous brain activity in early-onset bipolar disorder: a review of functional magnetic resonance imaging studies. *Journal of Affective Disorders* 311, 463–471. <https://doi.org/10.1016/j.jad.2022.05.055>.
- Cheng, W., Rolls, E.T., Qiu, J., Yang, D., Ruan, H., Wei, D., Zhao, L., Meng, J., Xie, P., Feng, J., 2018. Functional connectivity of the precuneus in unmedicated patients with depression. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging* 3 (12), 1040–1049. <https://doi.org/10.1016/j.bpsc.2018.07.008>.
- Cheung, J.C., Sorgi-Wilson, K.M., Ciesinski, N.K., McCloskey, M.S., 2024. Examining the relationship between subtypes of rumination and non-suicidal self-injury: A meta-analytic review. *Suicide and Life-Threatening Behavior* 54, 528–555. <https://doi.org/10.1111/sltb.13051>.
- Choe, A., Hwang, S., Kim, J., Park, K., Choi, J., Hong, S., 2014. Validity of the K-WAIS-IV short forms. *Korean Journal of Clinical Psychology* 33 (2), 413–428. <https://doi.org/10.15842/kjcp.2014.33.2.011>.
- Choi, Y., 2007. Assessing Emotion Dysregulation: Psychometric Properties of the Korean Version of the Difficulties in Emotion Regulation Scale. *Korean Journal of Clinical Psychology* 26 (4), 1015–1038. <https://doi.org/10.15842/kjcp.2007.26.4.012>.
- De Luca, L., Pastore, M., Palladino, B.E., Reime, B., Warth, P., Menesini, E., 2023. The development of Non-Suicidal Self-Injury (NSSI) during adolescence: A systematic review and Bayesian meta-analysis. *Journal of Affective Disorders*. <https://doi.org/10.1016/j.jad.2023.07.091>.
- Doucet, G.E., Bassett, D.S., Yao, N., Glahn, D.C., Frangou, S., 2017. The role of intrinsic brain functional connectivity in vulnerability and resilience to bipolar disorder. *American Journal of Psychiatry* 174 (12), 1214–1222. <https://doi.org/10.1176/appi.ajp.2017.17010095>.
- First, M.B., Williams, J.B., Karg, R.S., Spitzer, R.L., 2017. SCID-5-CV. In: *Intervista Clinica Strutturata per i Disturbi del DSM-5*. Raffaello Cortina Editore Milano, *Versione Per Il Clinico Ed Italiana a cura Di Andrea Fossati e Serena Borroni*.
- First, M.B., Yousif, L.H., Clarke, D.E., Wang, P.S., Gogtay, N., Appelbaum, P.S., 2022. DSM-5-TR: Overview of what's new and what's changed. *World Psychiatry* 21 (2), 218. <https://doi.org/10.1002/wps.20989>.
- Fortin, J.-P., Parker, D., Tunc, B., Watanabe, T., Elliott, M.A., Ruparel, K., Roalf, D.R., Satterthwaite, T.D., Gur, R.C., Gur, R.E., 2017. Harmonization of multi-site diffusion tensor imaging data. *Neuroimage* 161, 149–170. <https://doi.org/10.1016/j.neuroimage.2017.08.047>.
- Freedman, D., Lane, D., 1983. A nonstochastic interpretation of reported significance levels. *Journal of Business & Economic Statistics* 1, 292–298. <https://doi.org/10.1080/07350015.1983.10509354>.
- Grandclerc, S., De Labrouhe, D., Spodenkiewicz, M., Lachal, J., Moro, M.-R., 2016. Relations between nonsuicidal self-injury and suicidal behavior in adolescence: a systematic review. *PLoS One* 11 (4), e0153760. <https://doi.org/10.1371/journal.pone.0153760>.
- Gratz, K.L., Roemer, L., 2004. Multidimensional assessment of emotion regulation and dysregulation: Development, factor structure, and initial validation of the difficulties in emotion regulation scale. *Journal of Psychopathology and Behavioral Assessment* 26, 41–54. <https://doi.org/10.1023/B:JOBA.0000007455.08539.94>.
- Griffanti, L., Douaud, G., Bijsterbosch, J., Evangelisti, S., Alfaro-Almagro, F., Glasser, M.F., Duff, E.P., Fitzgibbon, S., Wehrl, R., Carone, D., Beckmann, C.F., Smith, S.M., 2017. Hand classification of fMRI ICA noise components. *Neuroimage* 154, 188–205. <https://doi.org/10.1016/j.neuroimage.2016.12.036>.
- Himberg, J., Hyvärinen, A., Esposito, F., 2004. Validating the independent components of neuroimaging time series via clustering and visualization. *Neuroimage* 22 (3), 1214–1222. <https://doi.org/10.1016/j.neuroimage.2004.03.027>.
- Ho, T.C., Walker, J.C., Teresi, G.L., Kulla, A., Kirshenbaum, J.S., Gifuni, A.J., Singh, M.K., Gotlib, I.H., 2021. Default mode and salience network alterations in suicidal and non-suicidal self-injurious thoughts and behaviors in adolescents with depression. *Translational Psychiatry* 11 (1), 38. <https://doi.org/10.1038/s41398-020-01103-x>.
- Johnson, W.E., Li, C., Rabinovic, A., 2007. Adjusting batch effects in microarray expression data using empirical Bayes methods. *Biostatistics* 8, 118–127. <https://doi.org/10.1093/biostatistics/kxj037>.
- Kaess, M., Hooley, J.M., Klimes-Dougan, B., Koenig, J., Plener, P.L., Reichl, C., Robinson, K., Schmah, C., Sicorello, M., Westlund Schreiner, M., Cullen, K.R., 2021. Advancing a temporal framework for understanding the biology of nonsuicidal self-injury: An expert review. *Neuroscience & Biobehavioral Reviews* 130, 228–239. <https://doi.org/10.1016/j.neubiorev.2021.08.022>.
- Kebets, V., Holmes, A.J., Orban, C., Tang, S., Li, J., Sun, N., Kong, R., Poldrack, R.A., Yeo, B.T., 2019. Somatosensory-motor dysconnectivity spans multiple transdiagnostic dimensions of psychopathology. *Biological Psychiatry* 86 (10), 779–791. <https://doi.org/10.1016/j.biopsych.2019.06.013>.
- Khodadadifar, T., Soltaninejad, Z., Ebneabbasi, A., Eickhoff, C.R., Sorg, C., Van Eimeren, T., Vogeley, K., Zarei, M., Eickhoff, S.B., Tahmasian, M., 2022. In search of convergent temporal brain abnormality in cognitive emotion regulation: a transdiagnostic neuroimaging meta-analysis. *Human Brain Mapping* 43 (4), 1309–1325. <https://doi.org/10.1002/hbm.25722>.
- Kieken, G., Hasking, P., Bruffaerts, R., Alonso, J., Auerbach, R.P., Bantjes, J., Benjet, C., Boyes, M., Chiu, W.T., Claes, L., 2023. Non-suicidal self-injury among first-year college students and its association with mental disorders: results from the World Mental Health International College Student (WMH-ICS) initiative. *Psychological Medicine* 53 (3), 875–886. <https://doi.org/10.1017/S0033291721002245>.
- Kim, J., Andrews-Hanna, J.R., Eisenbarth, H., Lux, B.K., Kim, H.J., Lee, E., Lindquist, M.A., Losin, E.A.R., Wager, T.D., Woo, C.-W., 2023. A dorsomedial prefrontal cortex-based dynamic functional connectivity model of rumination. *Nature Communications* 14, 3540. <https://doi.org/10.1038/s41467-023-39142-9>.
- Kim, S., Kim, Y., Hur, J.-W., 2019. Nonsuicidal self-injury among Korean young adults: a validation of the Korean version of the inventory of statements about self-injury. *Psychiatry Investigation* 16 (4), 270. <https://doi.org/10.30773/pi.2019.01.23>.
- Kiviniemi, V., Starck, T., Remes, J., Long, X., Nikkinen, J., Haapea, M., Veijola, J., Moilanen, L., Isohanni, M., Zang, Y.F., Tervonen, O., 2009. Functional segmentation of the brain cortex using high model order group PICA. *Human Brain Mapping* 30 (12), 3865–3886. <https://doi.org/10.1002/hbm.20813>.
- Klonsky, E.D., Glenn, C.R., 2009. Assessing the functions of non-suicidal self-injury: Psychometric properties of the Inventory of Statements About Self-Injury (ISAS). *Journal of Psychopathology and Behavioral Assessment* 31, 215–219. <https://doi.org/10.1007/s10862-008-9107-z>.
- Kohn, N., Eickhoff, S.B., Scheller, M., Laird, A.R., Fox, P.T., Habel, U., 2014. Neural network of cognitive emotion regulation—an ALE meta-analysis and MACM analysis. *Neuroimage* 87, 345–355. <https://doi.org/10.1016/j.neuroimage.2013.11.001>.
- Kropf, E., Syan, S.K., Minuzzi, L., Frey, B.N., 2018. From anatomy to function: the role of the somatosensory cortex in emotional regulation. *Brazilian Journal of Psychiatry* 41 (3), 261–269. <https://doi.org/10.1590/1516-4446-2018-0183>.
- Kuehn, K.S., Dora, J., Harned, M.S., Foster, K.T., Song, F., Smith, M.R., King, K.M., 2022. A meta-analysis on the affect regulation function of real-time self-injurious thoughts and behaviours. *Nature Human Behaviour* 6 (7), 964–974. <https://doi.org/10.1038/s41562-022-01340-8>.
- Kwon, S.-M., 1992. Differential roles of dysfunctional attitudes and automatic thoughts in depression: An integrated cognitive model of depression [Unpublished doctoral dissertation]. University of Queensland.
- Li, M., Xiao, Y., Ge, Y., Yan, H., Li, X., Yue, W., Yan, H., 2026. Functional heterogeneity in non-suicidal self-injury across psychiatric disorders: Neural and psychosocial correlates. *Translational Psychiatry* 16, 60. <https://doi.org/10.1038/s41398-025-03802-9>.
- Martino, M., Magioncalda, P., Huang, Z., Conio, B., Piaggio, N., Duncan, N.W., Rocchi, G., Escelsior, A., Marozzi, V., Wolff, A., 2016. Contrasting variability patterns in the default mode and sensorimotor networks balance in bipolar depression and mania. *Proceedings of the National Academy of Sciences* 113 (17), 4824–4829. <https://doi.org/10.1073/pnas.1517558113>.
- Muehlenkamp, J.J., Brausch, A.M., 2019. Protective factors do not moderate risk for past-year suicide attempts conferred by recent NSSI. *Journal of Affective Disorders* 245, 321–324. <https://doi.org/10.1016/j.jad.2018.11.013>.
- Muehlenkamp, J.J., Tillotson, V., 2024. 127/Oversight and Epidemiology of NSSI in Clinical Samples. In: Lloyd-Richardson, E.E., Baetens, I., Whitlock, J.L. (Eds.), *The Oxford Handbook of Nonsuicidal Self-Injury*. Oxford University Press, p. 0. <https://doi.org/10.1093/oxfordhb/9780197611272.013.8>.
- Muehlenkamp, J.J., Xhuma, N., Brausch, A.M., 2019. Self-injury age of onset: A risk factor for NSSI severity and suicidal behavior. *Archives of Suicide Research*. <https://doi.org/10.1080/13811118.2018.1486252>.
- Nagy, L.M., Shanahan, M.L., Seaford, S.P., 2023. Nonsuicidal selfinjury and rumination: A meta-analysis. *Journal of Clinical Psychology* 79, 7–27. <https://doi.org/10.1002/jclp.23394>.
- Niu, Z., Luo, H., Zhang, X., Wu, X., Tang, Q., Chen, C., Li, J., 2023. Brain-gut axis mechanism of subthreshold nonsuicidal self-injury addictive features in adolescents. *Cerebral Cortex* 33 (21), 10784–10792. <https://doi.org/10.1093/cercor/bhad322>.
- Nock, M.K., 2009. Why do people hurt themselves? New insights into the nature and functions of self-injury. *Current directions in psychological science* 18 (2), 78–83. <https://doi.org/10.1111/j.1467-8721.2009.01613.x>.

- Ochsner, K.N., Silvers, J.A., Buhle, J.T., 2012. Functional imaging studies of emotion regulation: A synthetic review and evolving model of the cognitive control of emotion. *Annals of the New York Academy of Sciences* 1251, E1–E24. <https://doi.org/10.1111/j.1749-6632.2012.06751.x>.
- Plener, P.L., Kaess, M., Schmahl, C., Pollak, S., Fegert, J.M., Brown, R.C., 2018. Nonsuicidal self-injury in adolescents. *Deutsches Ärzteblatt International* 115 (3), 23. <https://doi.org/10.3238/arztebl.2018.0023>.
- Power, J.D., Barnes, K.A., Snyder, A.Z., Schlaggar, B.L., Petersen, S.E., 2012. Spurious but systematic correlations in functional connectivity MRI networks arise from subject motion. *Neuroimage* 59 (3), 2142–2154. <https://doi.org/10.1016/j.neuroimage.2011.10.018>.
- Rodriguez, M., Kross, E., 2023. Sensory emotion regulation. *Trends in Cognitive Sciences* 27 (4), 379–390. <https://doi.org/10.1016/j.tics.2023.01.008>.
- Seo, D., Oltman, C.A., Haut, K.M., Sinha, R., MacDonald III, A.W., Patrick, C.J., 2014. Neural correlates of preparatory and regulatory control over positive and negative emotion. *Social Cognitive and Affective Neuroscience* 9 (4), 494–504. <https://doi.org/10.1093/scan/nst115>.
- Seo, S.S., Chung, U.-S., Rim, H.D., Jeong, S.H., 2009. Reliability and validity of the 20-item Toronto Alexithymia Scale in Korean adolescents. *Psychiatry Investigation* 6 (3), 173. <https://doi.org/10.4306/pi.2009.6.3.173>.
- Smith, S.M., Fox, P.T., Miller, K.L., Glahn, D.C., Fox, P.M., Mackay, C.E., Filippini, N., Watkins, K.E., Toro, R., Laird, A.R., 2009. Correspondence of the brain's functional architecture during activation and rest. *Proceedings of the National Academy of Sciences* 106, 13040–13045. <https://doi.org/10.1073/pnas.0905267106>.
- Turner, B.J., Porter, A.C., Robillard, C.L., 2024. Average ages of onset and time to transition between self-injurious thoughts and behaviors: retrospective evidence from two developmentally distinct samples. *Journal of Affective Disorders*. <https://doi.org/10.1016/j.jad.2024.07.105>.
- Watsky, R.E., Gotts, S.J., Berman, R.A., McAdams, H.M., Zhou, X., Greenstein, D., Lalonde, F.M., Gochman, P., Clasen, L.S., Shora, L., 2018. Attenuated resting-state functional connectivity in patients with childhood-and adult-onset schizophrenia. *Schizophrenia Research* 197, 219–225. <https://doi.org/10.1016/j.schres.2018.01.003>.
- Whitfield-Gabrieli, S., Nieto-Castanon, A., 2012. Conn: a functional connectivity toolbox for correlated and anticorrelated brain networks. *Brain Connectivity* 2 (3), 125–141. <https://doi.org/10.1089/brain.2012.0073>.
- Wolff, J.C., Thompson, E., Thomas, S.A., Nesi, J., Bettis, A.H., Ransford, B., Scopelliti, K., Frazier, E.A., Liu, R.T., 2019. Emotion dysregulation and non-suicidal self-injury: A systematic review and meta-analysis. *European Psychiatry* 59, 25–36. <https://doi.org/10.1016/j.eurpsy.2019.03.004>.
- Yook, S., Kim, Z., 1997. A clinical study on the Korean version of Beck Anxiety Inventory: Comparative study of patient and non-patient. *Korean Journal of Clinical Psychology* 16, 185–197.
- Yu, B.-K., Lee, H.-K., Lee, K.-S., 2011. Validation and factor structure of Korean version of the Beck Depression Inventory Second Edition (BDI-II): in a university student sample. *Korean Journal of Biological Psychiatry* 18 (3), 126–133.
- Zhang, X., Wu, B., Yang, X., Kemp, G.J., Wang, S., Gong, Q., 2024. Abnormal large-scale brain functional network dynamics in social anxiety disorder. *CNS neuroscience & therapeutics* 30 (8), e14904. <https://doi.org/10.1111/cns.14904>.
- Zhang, X., Yang, X., Wu, B., Pan, N., He, M., Wang, S., Kemp, G.J., Gong, Q., 2023. Large-scale brain functional network abnormalities in social anxiety disorder. *Psychological Medicine* 53 (13), 6194–6204. <https://doi.org/10.1017/S0033291722003439>.
- Zhang, X., Zhang, R., Lv, L., Qi, X., Shi, J., Xie, S., 2022. Correlation between cognitive deficits and dorsolateral prefrontal cortex functional connectivity in first-episode depression. *Journal of Affective Disorders* 312, 152–158. <https://doi.org/10.1016/j.jad.2022.06.024>.